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ORIGINAL INVESTIGATION

Importance of Various Training-Load Measures in Injury Incidence of Professional Rugby League Athletes

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Purpose: To investigate the ability of various internal and external training-load (TL) monitoring measures to predict injury incidence among positional groups in professional rugby league athletes. ***Methods:*** TL and injury data were collected across 3 seasons (2013–2015) from 25 players competing in National Rugby League competition. Daily TL data were included in the analysis, including session rating of perceived exertion (sRPE-TL), total distance (TD), high-speed-running distance (>5 m/s), and high-metabolic-power distance (HPD; >20 W/kg). Rolling sums were calculated, nontraining days were removed, and athletes' corresponding injury status was marked as "available" or "unavailable." Linear (generalized estimating equations) and nonlinear (random forest; RF) statistical methods were adopted. ***Results:*** Injury risk factors varied according to positional group. For adjustables, the TL variables associated most highly with injury were 7-d TD and 7-d HPD, whereas for hit-up forwards they were sRPE-TL ratio and 14-d TD. For outside backs, 21- and 28-d sRPE-TL were identified, and for wide-running forwards, sRPE-TL ratio. The individual RF models showed that the importance of the TL variables in injury incidence varied between athletes. ***Conclusions:*** Differences in risk factors were recognized between positional groups and individual athletes, likely due to varied physiological capacities and physical demands. Furthermore, these results suggest that robust machine-learning techniques can appropriately monitor injury risk in professional team-sport athletes.

Keywords: machine learning, injuries, team sport, GPS, random forests, monitoring

Professional rugby league athletes undergo a vigorous training regimen that aims to facilitate positive training adaptations and improvements in performance. Together with extensive match demands,[1] the workloads these athletes regularly undergo are associated with an increased likelihood of injury,[2,3] so it is important that careful planning and prescription of training load (TL) be achieved.[4] These TLs are manipulated according to specific training-cycle objectives, with a focus on achieving balance between the minimum TL required to elicit adaptations and the maximum tolerable TL that can be applied before increasing injury risk.[2] Although many of the injuries sustained are due to physical collisions and tackles,[5] a considerable number can be classified as noncontact, which reflect soft-tissue injuries that are likely caused by poor physical preparation and/or excessive TL coupled with inadequate recovery.[2,6]

In an attempt to effectively manage this training stress and prevent inappropriate TL, the range of athlete-workload data being collected and monitoring systems being implemented is increasing.[4,7] Measurement tools such as global positioning systems (GPS)[8] and self-reported measures (eg, session rating of perceived exertion [sRPE])[9] allow sport scientists to manage individual athletes' workloads. Despite the emphasis placed on these monitoring systems, more research is necessary to understand the load–injury relationship. However, the complexity of the data analysis required to examine this relationship often restricts such investigations. More-robust classification-based machine-learning algorithms that appropriately handle nonlinear relationships offer 1 such method to examine the multifaceted relationships.

The potential for the application of machine-learning techniques to investigate complex relationships has recently been highlighted,[10] and such approaches have recently been used to identify risk factors of self-reported illness of professional rugby league athletes.[11] Furthermore, the predictive capacity of a novel workload ratio and its relationship with injury incidence in professional rugby league players have recently been presented.[6] Results showed that the ratio of acute (7-d) to chronic (28-d) total distance (TD) exhibited predictive capacity, with a very high acute:chronic workload ratio (>2.11) related to greater risk of injury in the current week (16.7%) and subsequent week (11.8%). That research provided useful TL-threshold values that are related to injury incidence, permitting practitioners to prescribe workloads that do not expose athletes to increased injury risk. However, the research is limited, in that it did not incorporate indicators of high-intensity workloads.

While the TD covered by athletes provides a measure of the volume of work completed, it must be emphasized that rugby league training and competition are both high-intensity and intermittent in nature.[12] Specifically, these demands involve frequent bouts of high-speed running and high-intensity acceleration and deceleration efforts, often performed at low speeds,[13] and it is important that these efforts be appropriately quantified. The metabolic-power method[14] represents a theoretical model for quantifying the energetic cost of such high-intensity actions, and although the validity of this measure as a marker of energy consumption has been questioned,[15] the metric provides a stable indication of overall high-intensity load, complementing traditional velocity-based measures.[16] Together, these methods provide greater insight into the high-intensity physical demands of rugby league that are useful in the prescription of training and monitoring workloads. However, no study has yet examined the relationship with injury incidence.

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Currently, there is a lack of evidence describing the workload–injury relationship of rugby league athletes according to position, despite the varied physical and physiological attributes evident between these groups.[17] These positional groups have diverse match demands, where forwards are exposed to a greater number of heavy collisions while backs cover greater distances at higher speeds with fewer collisions.[12] In addition, no study has examined the importance of TL to injury risk at the individual level, despite athletes' varied responses to training.[18] Thus, the physical demands and subsequent responses of individual athletes are varied and should be monitored accordingly.

The purpose of this study was to identify TL measures that possess the greatest association with injury risk for professional rugby league athletes. First, position-specific analyses identify TL variables that are most related to injury status. Next, machine-learning models demonstrate the importance of these variables in understanding injury at an individual level, where the results may be implemented in routine monitoring practices.

Methods

Subjects

Twenty-five professional rugby league players from 1 National Rugby League (NRL) club were recruited for this study. Of these players, 20 (80%) played 3 seasons, 3 (12%) played 2 seasons, and 2 (8%) played 1 season, overall giving a total of 68 player seasons. Players were categorized into 4 positional groups including hit-up forwards (HUF; props and locks, n = 6), wide-running forwards (WRF; second rowers, n = 5), adjustables (hookers, halfbacks, five-eighths, and fullbacks, n = 8), and outside backs (OB; centers and wingers, n = 6).[12]

Design

A longitudinal research design was conducted, where TL and injury data were collected during the entire 2013, 2014, and 2015 Australian NRL seasons. These data were used to develop linear and nonlinear (machine-learning) models that were able to identify TL variables associated with injury incidence. All participants provided informed consent before the commencement of the study. The institutional human ethics committee approved all experimental procedures.

Methodology

Quantifying Workloads. The intensity of all training sessions (field, resistance, and other sessions) and matches was provided by the athletes using the modified Borg CR-10 RPE scale.[19] This intensity rating was then multiplied by the duration of the session (min), providing a valid and reliable indication of internal training load (sRPE-TL).[20] Movement demands for field sessions and matches were quantified using GPS units, which used a raw sampling rate of 5 Hz that was interpolated to 15 Hz (SPI HPU, GPSports, Canberra, Australia). Each session was downloaded and trimmed so that only data obtained during drills or match play were retained for analysis, using the appropriate proprietary software (Team AMS, GPSports, Canberra, Australia). These units exhibit acceptable reliability for the quantification of high-intensity team-sport movement demands.[16] In addition to TD covered (m), distances over predetermined high-speed-running (HSD; >5 m/s) and high-metabolic-power (HPD; >20 W/kg) thresholds were chosen as indicators of high-intensity-running load, which are deemed as possessing acceptable reliability when detecting high-intensity movements.[16]

Injury Definition. For the purpose of the study, an injury was defined as any time-loss injury that did not result from a collision and that caused the player to be unable to complete full training or to miss match time.[21] Therefore, these injuries included all "preventable" soft-tissue overuse injuries. If an injury was present the player was labeled "unavailable," and injury-free days were labeled "available." Athletes were included in the data set if they sustained more than 3 injuries in total. All injuries were recorded and treated by medical staff.

Data Analysis. Daily data were included in the analysis; the corresponding TL values and injury status were recorded for each player. The daily data began on the first day of training until the last of that particular season, allowing chronological rolling loads to be calculated for each variable included, using 7-, 14-, 21-, and 28-day values. In addition, the acute:chronic ratio was calculated for each measure, using the 7- and 28-day rolling values.[6] These rolling loads were included rather than 7-day blocks, as team-sport athletes experience varied match-recovery periods,[22] wherein with a shorter (5-d) match-recovery period, 2 matches may be performed in a week, causing TL to spike. Days where no training was undertaken were removed from the analysis, as this represents no injury risk to the athlete, but the rolling-load calculations are inclusive of the recovery days. Overall, the percentage of available marked days was 97.5%, and 2.5% were marked unavailable.

Statistical Analyses

Generalized estimating equations (GEE) models were used to examine the relationship of each of the TL measures and injury status. This regression-based technique is used in generalized linear modeling, which is useful when there are unknown correlations between data and repeated observations of individuals. A Gaussian distribution was assumed, and an unstructured correlation structure was used. Separate models were constructed for each positional group to more specifically identify the TL variables that were most associated with injury status (available or unavailable). The parameter estimates were then examined to select the 5 most important TL variables of each positional group, as individual-athlete predictive models were subsequently developed based on these results. The model goodness of fit was assessed using the quasi likelihood under independence model criterion (QIC), and parameter estimates were described using Wald chi-squared (χ^2).

Next, machine-learning (random forest; RF) models were developed for each athlete to demonstrate the importance of the TL measures in contributing to injury incidence. The RF algorithm is a nonlinear categorical method that is an extension of the decision-tree approach. It is considered robust to noise (ie, variables that have little relationship with the target variable)[23] such as that associated with an injury-status variable. Before the development of these models, the data were partitioned into training, validation, and testing data sets (70/15/15%). The models were constructed using the training set, then tested in "unseen" validation data, providing a true assessment of performance. Although this method can be used to predict binary outcomes, for the current study the information provided from the models was used to demonstrate the individual variance in importance of predictor variables. This importance is determined by the number of times that a variable appears in the ensemble of decision trees, where the greater the value in the importance

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plot, the stronger the association with injury incidence, displayed relative (0–1) per athlete. The performances of these models were screened using receiver-operating-characteristic (ROC) curves, providing information regarding the sensitivity (true positive rate) by 1 minus specificity (false positive rate). The area under the ROC curve indicates the performance of the model, where a value of 1 indicates 100% accuracy in predicting the target variable.[23] All statistical analyses were conducted R statistical software (R.2.1, R Foundation for Statistical Computing).

Results

A total of 156 new injuries (athlete being unavailable) were recorded. The GEE models identified the TL measures that were most highly associated with injury for each positional group. Specifically, for the ADJ group (in order of importance), the TL variables recognized were 7-day TD, 7-day HPD, HPD ratio, 28-day HSD, and 21-day HPD, where the model QIC was 566.5 and the statistical significance of these measures ranged from $P = .001$ to .091. For the HUF group, identified variables were sRPE-TL ratio, 14-day TD, 14-day HPD, 7-day sRPE-TL, and 28-day HPD, where the QIC was 441.7 and significance of these measures ranged from $P = .006$ to .138. The OB models indicated that 21- and 28-day sRPE-TL, 7-day HSD, HPD ratio, and TD ratio were most associated with injury risk, with a QIC of 406.6 and statistical significance of these measures ranging from $P = .092$ to .225. For the WRF group, sRPE-TL ratio, HSD ratio, 14-day HPD, 14-day TD, and 7-day sRPE-TL were the most likely contributors to injury risk, with a QIC of 410.5 and statistical significance ranging from $P = .068$ to .830.

Results from the RF models are shown in Figure 1, which shows that the relative importance of each TL measure varied for each player. Specifically, for the ADJ group, the relative importance of TL variables was similar, with the exception of player 2, whose HPD ratio was rated lower than that of other players. The mean (± SD) ROC of the RF models was 0.74 ± 0.24. For the HUF group, 7-day sRPE-TL and 14-day HPD were associated the greatest importance in injury, where the mean model ROC was 0.65 ± 0.06. Variables recognized for the WRF group varied substantially between athletes, and for 2 players in particular (1 and 2). The mean model ROC was 0.64 ± 0.05. Similarly for the OB group, the importance of TL variables varied for players 3 and 6, showing a large discrepancy for TD ratio and 28-day sRPE-TL. These models had a mean ROC of 0.64 ± 0.04.

Discussion

The aim of this study was to examine the position- and athlete-level importance of various TL measures in injury risk by adopting linear and nonlinear statistical techniques. Results showed that the TL measures that were associated with injury differed between positional groups, and, when examined individually, the relative importance of the variables also varied. Collectively, these results provide a useful framework for sport scientists to more effectively individualize athlete workloads, given the known disparities in physical requirements of match play between rugby league positional groups[24] and, furthermore, the differences in physiological capacities of such athletes.[25]

TL monitoring is essential in ensuring that athletes are exposed to an adequate training stimulus while remaining within their capable limits before an increase in injury risk occurs.[4] In team sports such as rugby league, there has been a lack of research determining which of the commonly employed TL measures are useful in understanding injury risk using an individualized approach. The results of the present research demonstrate that both internal- and external-TL measures often collected in team sports can provide insight into injury risk, but it is important that these data be correctly analyzed and interpreted. As such, this study provides a more robust understanding and methodology of the complex etiology surrounding injury incidence.

In the context of rugby league, it is important to monitor TL not only at a positional level but also individually, due to the vast difference in movement demands of rugby league positional groups.[13,24] As the physical and physiological profiles of these athletes differ,[26] individualized monitoring approaches should be applied. As such, a novel component of this research was the individualized machine-learning models to understand the complex etiology of injury incidence. A similar approach was recently used in understanding relationships between internal and external TL in professional Australian football players.[27] Together, these provide evidence for the use of machine-learning techniques in monitoring team-sport athletes' workloads, but more research is necessary to improve model performance. Potentially, athlete-monitoring databases or software may incorporate such algorithms into their systems and further use methods such as deep learning.

Variables that were recognized as most highly associated with injury risk for all positional groups included 7-day sRPE-TL, 7-day TD, HPD ratio, and sRPE-TL ratio. Despite this, there was a variation in the importance of risk factors according to positional group. For example, for the ADJ group the TL variables associated most highly with injury were 7-day TD and 7-day HPD. This reflects the considerable acceleration-based demands of this position, where players are regularly required to perform high-intensity acceleration and deceleration efforts, often at low speeds.[13,24] For the HUF group, sRPE-TL ratio and 14-day TD were associated highly with injury risk; HUFs are most involved in heavy collisions throughout a match[12] and thus are restricted in covering greater distances. This indicates that changes in perceived intensity and being exposed to fluctuations in the volume of running suggest injury risk. For OBs the 21- and 28-day sRPE-TL were associated most highly with risk. This group covers considerable distances at higher speeds than all other positional groups[12]; our results suggest that changes in chronic internal TL may indicate elevated injury risk. The WRF group showed that sRPE-TL ratio and HSD ratio were associated with injury status; during match play, WRFs are exposed to a large number of collisions interspersed with a great amount of repeated high-intensity acceleration/deceleration efforts.[12] These results suggest that changes in acute:chronic perceived intensity and HSD should be monitored to reduce injury incidence.

The individual RF models showed that the importance of the TL variables in injury incidence varied between athletes. Individual responses to training vary,[18] and athletes' ability to tolerate TL depends on their capacity, characterized by their physiological profile, training status, and potentially other factors.[28] To account for this, the current study adopted an individualized approach to understand risk factors, demonstrating that despite the positional group of the athlete, the relative risk factors differed. While more effort is required to improve the individual-athlete model performances (described by the ROC value), the ever-increasing quantity of athlete-monitoring data collected may provide more insight into the complex relationship surrounding injury incidence.

Collectively, these results demonstrate that it is important to accurately examine acute TL in addition to chronic or acute:chronic loads for all measures at an individual level. Furthermore, using the

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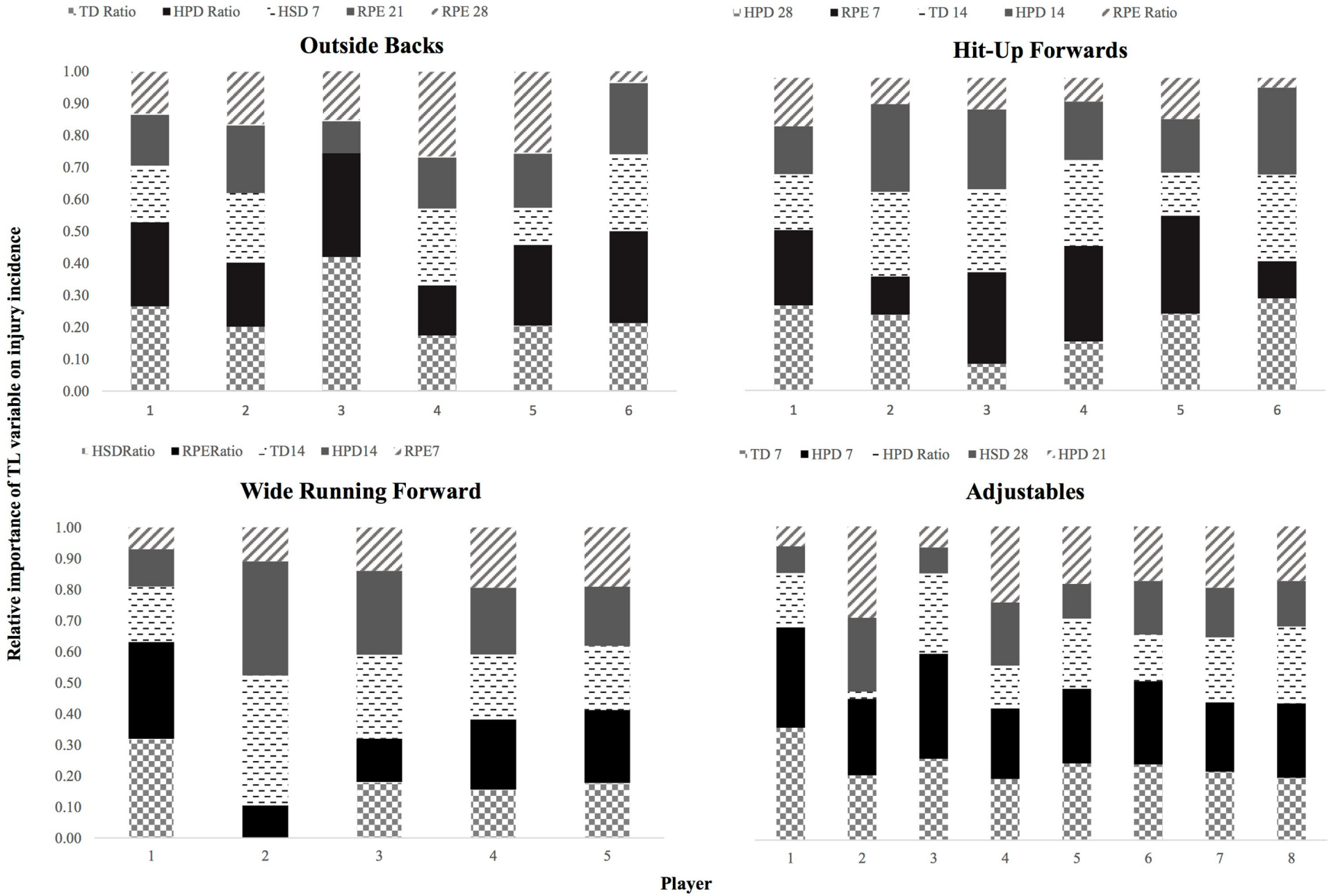

Figure 1 — Relative importance of training-load (TL) variables on individual athletes' injury risk using the random-forest algorithm. Abbreviations: TD, total distance; HPD, high-metabolic-power distance; HSD, high-speed-running distance; RPE, rating of perceived exertion.

rolling-load method rather than raw values in a specific block period is supported by this research. These results demonstrate that monitoring the shorter-term fluctuations in individuals' TL is essential, as this load has previously been suggested as the "fatigue" state of the athlete.[6] However, it must be noted that the results from this study are indicative of the injury risks for the population studied, so to understand other athletes' injury risk, these methods could be adopted. In addition, future research could examine the actual TL thresholds that are associated with injury risk, as the current study did not report on these.

Practical Implications

Using novel methodology, the outcomes of this research provide sport scientists with valid TL indicators of injury risk, specific to positional group, but also at an individual level. This reflects the large disparity in the movement demands across positional groups[12] and, thus, physiological profiles of athletes. Furthermore, this research provided further evidence for the use of machine-learning methods to provide information about the importance of various workload measures (ie, HPD, HSD) that are increasingly used in team sports such as rugby league. Collectively, these results assist in individualized training prescription and monitoring by aiding understanding of the variables that best predict injury risk on a positional and individual basis. Such analyses may help athletes avoid being exposed to an increased risk of injury.

Conclusion

These findings provide a framework for sport scientists to appropriately manage and monitor individual-athlete TLs using more comprehensive machine-learning techniques. Overall, it was recognized that both internal- and external-TL measures were associated with injury; the importance of these measures varied considerably between positional groups and individual athletes. This may be attributed to the known disparities in physical requirements of match play between these groups[24] and thus the physiological capacities of individual athletes.[25] Practically, this assists in the prescription and monitoring of training by making more-informed decisions and minimizing the risk of injury that is debilitating to performance.

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